

Math Bootcamp

Day 4 — Optimization

Math Camp 2026

Day 4 roadmap

1. **4.1** Unconstrained optimization — FOC and SOC
2. **4.2** Concavity, convexity, quasiconcavity — global optimality
3. **4.3** Equality constraints — Lagrange multipliers
4. **4.4** Envelope theorem — value functions
5. **4.5** Inequality constraints — Farkas lemma and KKT
6. **4.6** Why equality multipliers are unrestricted

4.1 The optimization problem

Consider: $\max_{x \in (a,b)} f(x)$ where f is differentiable.

Goal: Find conditions guaranteeing x^* is a local maximum.

Two types of conditions:

- **Necessary:** Must hold at any local max, but doesn't guarantee it
- **Sufficient:** If they hold, x^* is definitely a local max

4.1 First-Order Condition (necessary)

Theorem. If x^* is an interior local maximum and f is differentiable at x^* , then $f'(x^*) = 0$.

Proof.

- If $f'(x^*) > 0$: then $f(x^* + h) > f(x^*)$ for small $h > 0$, contradiction
- If $f'(x^*) < 0$: then $f(x^* + h) > f(x^*)$ for small $h < 0$, contradiction

A point with $f'(x^*) = 0$ is called a **critical point**.

4.1 Critical points are not always maxima

FOC finds critical points, but these could be:

- Local maxima
- Local minima
- Saddle points (inflection)

Example: $f(x) = x^3$ has $f'(x) = 3x^2 = 0$ at $x^* = 0$, but this is an inflection point, not a maximum.

Need additional conditions to distinguish maxima from other critical points.

4.1 Exercise: Finding critical points Together

Exercise. Find all critical points of $f(x) = x^3 - 3x$.

Solution.

- $f'(x) = 3x^2 - 3 = 3(x^2 - 1) = 0$ when $x = \pm 1$
- At $x^* = 1$: $f(1) = -2$, nearby values $f(0.9) = -2.079$, $f(1.1) = -2.069$
- This is a **local minimum**
- At $x^* = -1$: $f(-1) = 2$, nearby values show this is a **local maximum**

FOC alone cannot distinguish max from min — need SOC.

4.1 Second-Order Condition (sufficient)

Theorem. If $f'(x^*) = 0$ and $f''(x^*) < 0$, then x^* is a strict local maximum.

Proof sketch. Taylor expansion around x^* :

$$f(x^* + h) = f(x^*) + \underbrace{f'(x^*)}_{=0} h + \frac{1}{2} f''(x^*) h^2 + o(h^2)$$

- For small h , the h^2 term dominates the remainder
- Since $f''(x^*) < 0$, we have $f(x^* + h) < f(x^*)$ for small $h \neq 0$

Similarly, $f''(x^*) > 0$ implies strict local minimum.

4.1 The gap: When is FOC alone sufficient?

SOC requires checking second derivatives at each critical point.

Question: Can we guarantee a maximum using only FOC, without checking SOC everywhere?

Answer: Yes, if f has the right **global shape** (concavity).

Economic intuition: Profit maximization with decreasing returns — any critical point is the global maximum.

This leads us to study **concavity** and **convexity**.

4.1 Exercise: Testing SOC Together

Exercise. For $f(x) = \ln(x) - x$ on $x > 0$, find critical points and verify SOC.

Solution.

- $f'(x) = \frac{1}{x} - 1 = 0$ gives $x^* = 1$
- $f''(x) = -\frac{1}{x^2}$
- At $x^* = 1$: $f''(1) = -1 < 0 \Rightarrow$ strict local maximum

Since $f''(x) = -\frac{1}{x^2} < 0$ for all $x > 0$, the function is strictly concave, so $x^* = 1$ is the unique global maximum.

4.1 Important caveat: Strict concavity vs. $f'' < 0$

Warning. Strict concavity does **not** require $f''(x) < 0$ everywhere.

Example: $f(x) = -x^4$ is strictly concave on $\mathbb{R}$, but $f''(0) = 0$.

At $x^* = 0$:

- $f'(0) = 0$ (critical point)
- $f''(0) = 0$ (SOC with strict inequality fails)
- Yet $f(x) = -x^4 < 0 = f(0)$ for all $x \neq 0$ (strict local max)

Lesson: $f''(x) < 0$ is sufficient but not necessary for strict concavity. The condition $f''(x) \leq 0$ is necessary and sufficient for (non-strict) concavity.

4.2 Global shape and global optima

FOC/SOC give local information. For global optimality, we need global shape.

Key insight:

- **Concave functions:** Any local maximum is a global maximum
- **Strictly concave:** The maximum is unique
- **Convex functions:** Any local minimum is a global minimum

Quasiconcavity: A weaker condition — FOC still gives global max on convex sets.

4.2 Concave functions: Definition

Definition. A function f is **concave** if for all x, y and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y).$$

Geometric interpretation:

- The function lies **above** its chords (secant lines)
- Epigraph $\{(x, t) : t \leq f(x)\}$ is a convex set

4.2 Tangent characterization

For differentiable f , an equivalent characterization:

$$f(y) \leq f(x) + f'(x)(y - x) \quad \text{for all } x, y.$$

Interpretation:

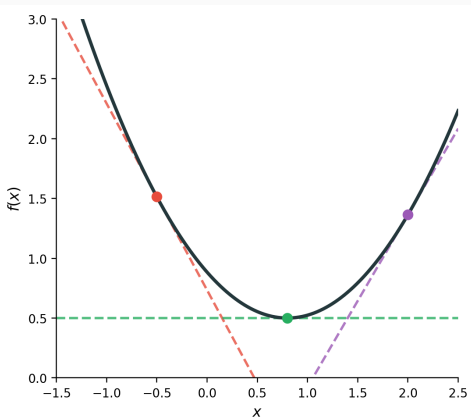
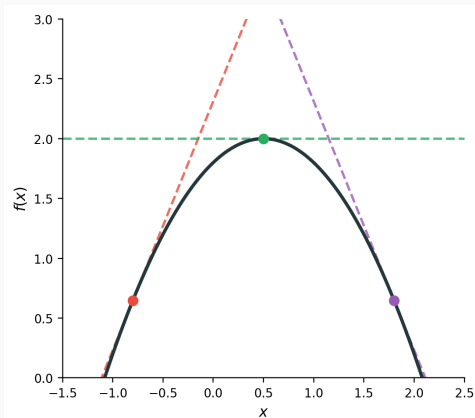
- The function lies **below** its tangent lines
- First-order Taylor approximation **overestimates** f

At a critical point where $f'(x^*) = 0$:

$$f(y) \leq f(x^*) + 0 \cdot (y - x^*) = f(x^*) \quad \text{for all } y.$$

So x^* is a **global** maximum!

4.2 Concave and convex figure



4.2 Second derivative and concavity

Theorem. For $f \in C^2$: f is concave $\Leftrightarrow f''(x) \leq 0$ for all x .

Proof ($\Rightarrow$: concave $\Rightarrow f'' \leq 0$).

From concavity: $f(x+h) + f(x-h) \leq 2f(x)$ for all $h > 0$.

Rearrange and divide by $h > 0$:

$$\frac{f(x+h) - f(x)}{h} \leq \frac{f(x) - f(x-h)}{h}.$$

Bring to one side and divide by h again:

$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} \leq 0.$$

Let $h \rightarrow 0$: limit is $f''(x) \leq 0$.

See Appendix A.1 for rigorous version using Taylor expansion

4.2 Proof: $f'' \leq 0$ implies concave

Proof ($\Leftarrow$: $f'' \leq 0 \Rightarrow$ concave).

Fix any x and y . Taylor around x :

$$f(y) = f(x) + f'(x)(y - x) + \frac{1}{2}f''(\xi)(y - x)^2,$$

where ξ lies between x and y .

- Since $f''(\xi) \leq 0$ and $(y - x)^2 \geq 0$, the second-order term is ≤ 0
- Therefore: $f(y) \leq f(x) + f'(x)(y - x)$
- This is the tangent characterization of concavity

Conclusion: $f'' \leq 0$ everywhere is necessary and sufficient for concavity.

4.2 Strict concavity

Definition. f is **strictly concave** if for all $x \neq y$ and $\lambda \in (0, 1)$:

$$f(\lambda x + (1 - \lambda)y) > \lambda f(x) + (1 - \lambda)f(y).$$

Second-order sufficient (not necessary).

- If $f''(x) < 0$ for all x (strictly negative), then f is strictly concave.
- But strict concavity does not require $f'' < 0$ everywhere.

Example. $f(x) = -x^4$ on $[-1, 1]$:

- Strictly concave (strict inequality holds)
- But $f''(0) = 0$ at $x = 0$

This shows $f'' < 0$ is sufficient but not necessary for strict concavity.

4.2 Exercise: Concavity implies global max Together

Exercise. Prove that if f is concave and $f'(x^*) = 0$, then x^* is a global maximum.

Solution.

- By tangent characterization: $f(y) \leq f(x^*) + f'(x^*)(y - x^*)$ for all y
- Since $f'(x^*) = 0$: $f(y) \leq f(x^*)$ for all y
- Therefore x^* is a global maximum

This bridges FOC (necessary) + concavity (sufficient) $\Rightarrow$ global optimality.

4.2 Convex functions

Definition. f is **convex** if for all x, y and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Tangent characterization: $f(y) \geq f(x) + f'(x)(y - x)$.

- Tangent lines lie below the graph
- First-order Taylor **underestimates** f

For convex f : if $f'(x^*) = 0$, then x^* is a global **minimum**.

f convex $\Leftrightarrow f''(x) \geq 0$ for C^2 functions.

4.2 Exercise: Testing concavity vs. convexity Together

Exercise. Determine if these functions are concave, convex, or neither:

(a) $f(x) = -x^2$ on $\mathbb{R}$

- $f''(x) = -2 < 0$ for all $x \Rightarrow$ **strictly concave**

(b) $f(x) = e^x$ on $\mathbb{R}$

- $f''(x) = e^x > 0$ for all $x \Rightarrow$ **strictly convex**

(c) $f(x) = x^3$ on $\mathbb{R}$

- $f''(x) = 6x$ changes sign at $x = 0 \Rightarrow$ **neither** concave nor convex on all of $\mathbb{R}$

4.2 Multivariate concavity and tangent planes

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the tangent **plane** at $\mathbf{x}$ is:

$$T(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}).$$

Concave (multivariate): $f(\mathbf{y}) \leq T(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y}$.

- Function lies below its tangent **hyperplane**
- In 2D: surface below tangent plane
- In 3D visualization: bowl shape opening downward

Hessian characterization: For $f \in C^2$, f is concave iff $H_f(\mathbf{x})$ is negative semidefinite for all $\mathbf{x}$.

4.2 Quasiconcave functions: Two definitions

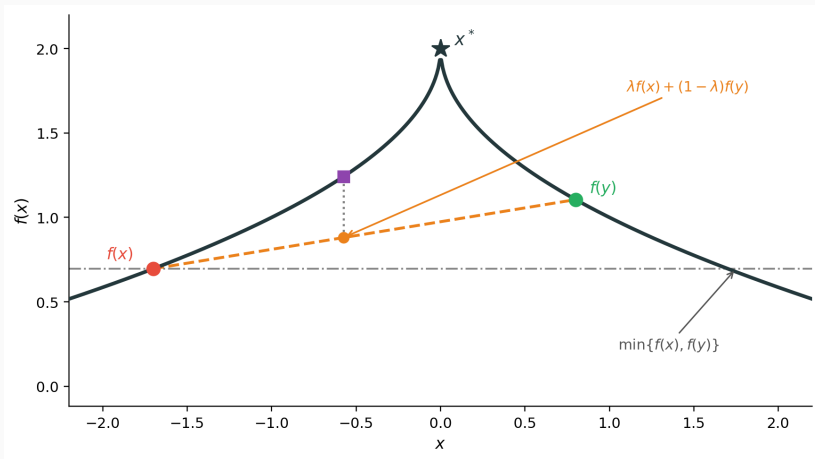
Definition A (segment). f is **quasiconcave** if for all x, y and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}.$$

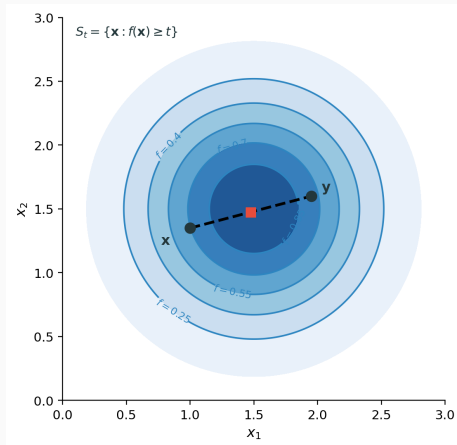
Along any segment, f stays at least as high as the lower endpoint.

Definition B (upper contours). f is quasiconcave if upper contour sets $S_t = \{\mathbf{x} : f(\mathbf{x}) \geq t\}$ are convex for all t .

4.2 One-dimensional quasiconcave function



4.2 Two-dimensional upper contour sets



4.2 Concave implies quasiconcave

Theorem. Concave $\Rightarrow$ quasiconcave, but not conversely.

Proof. For concave f and any $x, y, \lambda \in [0, 1]$:

- By concavity: $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$
- WLOG $f(x) \geq f(y)$, so $\min\{f(x), f(y)\} = f(y)$
- Then: $\lambda f(x) + (1 - \lambda)f(y) \geq f(y) = \min\{f(x), f(y)\}$
- Therefore: $f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$

This is the definition of quasiconcavity. The converse fails: e.g., sigmoid is quasiconcave but not concave.

4.2 Equivalence: segment vs. upper contours

Theorem. Definitions A and B are equivalent.

A: $f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$.

B: $S_t = \{\mathbf{x} : f(\mathbf{x}) \geq t\}$ is convex for every t .

Proof (A $\Rightarrow$ B). Let $\mathbf{x}, \mathbf{y} \in S_t$.

- Then $f(\mathbf{x}) \geq t, f(\mathbf{y}) \geq t$
- For $\lambda \in [0, 1]$: $f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\} \geq t$
- The segment stays in S_t , so S_t is convex

Proof (B $\Rightarrow$ A). WLOG $f(\mathbf{x}) \geq f(\mathbf{y})$.

- Then $\mathbf{x}, \mathbf{y} \in S_{f(\mathbf{y})}$
- By convexity of $S_{f(\mathbf{y})}$: the segment is in $S_{f(\mathbf{y})}$
- So $f(\text{segment}) \geq f(\mathbf{y}) = \min\{f(\mathbf{x}), f(\mathbf{y})\}$

4.2 Quasiconvex functions

Definition. f is **quasiconvex** if for all x, y and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\}.$$

Equivalently: lower contour sets $\{\mathbf{x} : f(\mathbf{x}) \leq t\}$ are convex.

Key relationship: f quasiconcave $\Leftrightarrow -f$ quasiconvex.

Example: $f(x) = x^2$ is quasiconvex (lower sets $[-a, a]$ are convex).

4.2 Log-concave and hierarchy

Definition. f is **log-concave** if $\ln f$ is concave (for $f > 0$).

Hierarchy (strongest to weakest):

1. Concave
2. Log-concave
3. Quasiconcave

Relationships:

- Concave and positive $\Rightarrow$ log-concave
- Log-concave $\Rightarrow$ quasiconcave

See Appendix A.2 for proofs

4.2 Economic examples

Cobb–Douglas: $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$

- Concave if $\alpha + (1 - \alpha) \leq 1$ (always, with equality)
- Quasiconcave for all $\alpha \in (0, 1)$

CES: $u = (x_1^\rho + x_2^\rho)^{1/\rho}$

- Log-concave for certain parameter ranges
- Quasiconcave more generally

Why this matters: FOC gives global max on convex budget set.

4.2 Why quasiconcavity matters

Theorem. If f is quasiconcave and the feasible set $\mathcal{B}$ is convex, then any feasible point with no first-order improving direction is a global maximum.

Intuition:

- Quasiconcavity ensures “no valleys” in the upper direction
- With convex feasible set, any local peak is the global peak

Cobb–Douglas is quasiconcave, so the FOC candidate gives the global max on a convex budget set.

4.2 Exercise: Cobb–Douglas quasiconcavity Together

Exercise. Show $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ is quasiconcave on $\mathbb{R}_{++}^2$.

Solution. Check that upper contour sets are convex:

- $\{u \geq t\}$ means $x_1^\alpha x_2^{1-\alpha} \geq t$
- Taking logs: $\alpha \ln x_1 + (1 - \alpha) \ln x_2 \geq \ln t$
- In $(\ln x_1, \ln x_2)$ coordinates: this is a half-space
- Half-spaces are convex
- The exponential map preserves convexity

Therefore S_t is convex, so u is quasiconcave.

4.2 Exercise: Cobb–Douglas is not concave Together

Exercise. Show $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ is **not** concave for $\alpha \in (0, 1)$.

Solution.

- The Hessian has mixed eigenvalue signs
- Along the ray $x_1 = x_2 = t$: $u = t$, which is linear (not strictly concave)
- The function is homogeneous of degree 1, so it cannot be strictly concave

This shows quasiconcavity is genuinely **weaker** than concavity.

4.2 Exercise: Testing quasiconcavity Take home

Exercise. Determine if $f(x, y) = xy$ is quasiconcave on $\mathbb{R}_{++}^2$.

Solution. Check upper contour sets $\{xy \geq t\}$.

- For $t > 0$: this is $\{(x, y) : y \geq t/x\}$ where $x > 0$
- Take $(1, t)$ and $(t, 1)$: both in the set since $1 \cdot t = t$
- Midpoint: $(\frac{1+t}{2}, \frac{1+t}{2})$ has product $(\frac{1+t}{2})^2$
- By AM-GM: $(\frac{1+t}{2})^2 \geq t$, so midpoint in set

The upper contour sets are convex, so $f(x, y) = xy$ is quasiconcave.

4.3 Constrained optimization problem

Problem. Maximize $f(\mathbf{x})$ subject to $h(\mathbf{x}) = 0$.

The constraint defines a curve (or surface) in the domain.

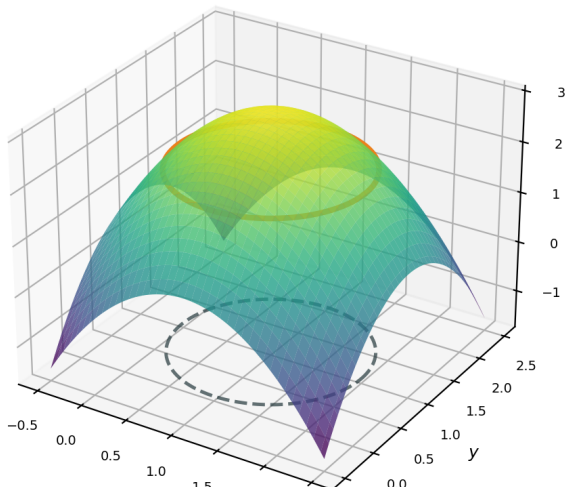
We seek the highest point of f on this surface.

Economic example. Maximize utility $u(x_1, x_2)$ subject to budget $p_1x_1 + p_2x_2 = w$.

Key insight: We cannot simply set $\nabla f = 0$ — must respect the constraint.

4.3 3D visualization

$$f = -(x-1)^2 - (y-1)^2 + 3 \text{ with } h(x,y) = (x-1)^2 + (y-1)^2 - 1 = 0$$



4.3 The tangent argument

At a constrained optimum $\mathbf{x}^*$ on the constraint surface:

- Any feasible direction $\mathbf{d}$ (tangent to constraint) satisfies $\nabla h(\mathbf{x}^*) \cdot \mathbf{d} = 0$
- If $\nabla f(\mathbf{x}^*) \cdot \mathbf{d} \neq 0$, we could move along constraint to increase f
- At optimum, no feasible improving direction exists

Conclusion. $\nabla f(\mathbf{x}^*)$ must be parallel to $\nabla h(\mathbf{x}^*)$.

That is: $\nabla f(\mathbf{x}^*) = -\mu \nabla h(\mathbf{x}^*)$ for some scalar μ .

4.3 Lagrangian and first-order conditions

Theorem. At a constrained optimum (with regularity), there exists $\mu \in \mathbb{R}$ such that

$$\nabla f(\mathbf{x}^*) + \mu \nabla h(\mathbf{x}^*) = \mathbf{0}, \quad h(\mathbf{x}^*) = 0.$$

Lagrangian. $\mathcal{L}(\mathbf{x}, \mu) = f(\mathbf{x}) + \mu h(\mathbf{x})$.

- Stationarity: $\nabla_{\mathbf{x}} \mathcal{L} = \nabla f + \mu \nabla h = \mathbf{0}$
- Constraint: $\partial \mathcal{L} / \partial \mu = h = 0$

μ is the **Lagrange multiplier** (shadow price of relaxing the constraint).

4.3 Exercise: Cobb–Douglas with budget Together

Exercise. Maximize $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ subject to $p_1 x_1 + p_2 x_2 = w$.

Solution.

- Lagrangian: $\mathcal{L} = u + \mu(w - p_1 x_1 - p_2 x_2)$
- FOCs: $\frac{\partial u}{\partial x_1} = \mu p_1$, $\frac{\partial u}{\partial x_2} = \mu p_2$
- Expanding: $\frac{\alpha u}{x_1} = \mu p_1$, $\frac{(1-\alpha)u}{x_2} = \mu p_2$
- Taking ratio: $\frac{\alpha x_2}{(1-\alpha)x_1} = \frac{p_1}{p_2}$

With budget: $p_1 x_1 = \alpha w$, $p_2 x_2 = (1 - \alpha)w$ (constant expenditure shares, Day 1).

4.3 Exercise: Cost minimization Together

Exercise. Minimize $C(q_1, q_2) = r_1 q_1 + r_2 q_2$ subject to $F(q_1, q_2) = \bar{y}$.

Solution.

- Lagrangian: $\mathcal{L} = r_1 q_1 + r_2 q_2 + \mu(\bar{y} - F(q_1, q_2))$
- FOCs: $r_1 = \mu F_1$, $r_2 = \mu F_2$ where $F_i = \frac{\partial F}{\partial q_i}$
- Therefore: $\frac{r_1}{r_2} = \frac{F_1}{F_2}$

The input price ratio equals the marginal rate of technical substitution.

μ is the marginal cost of producing additional output.

4.3 Exercise: Interpreting the multiplier Together

Exercise. In the Cobb–Douglas solution, what is the interpretation of μ ?

Solution.

- From FOC: $\mu = \frac{\alpha u}{p_1 x_1} = \frac{u}{w}$
- Substituting the solution: $u = \left(\frac{\alpha w}{p_1}\right)^\alpha \left(\frac{(1-\alpha)w}{p_2}\right)^{1-\alpha}$
- Thus: $\mu = \frac{\alpha^\alpha (1-\alpha)^{1-\alpha}}{p_1^\alpha p_2^{1-\alpha}}$

Since $\mu = \frac{\partial \mathcal{L}}{\partial w}$, this is the marginal utility of income — how much extra utility from one more dollar.

4.4 Value functions and parameters

Economic problems depend on parameters: prices, income, technology.

Value function. $V(\theta) = \max_{\mathbf{x}} f(\mathbf{x}; \theta)$ subject to constraints.

We want to know: how does the optimal value change when θ shifts?

Key question: Do we need to account for how $\mathbf{x}^*(\theta)$ changes?

4.4 Unconstrained envelope theorem

Theorem. For $V(\theta) = \max_{\mathbf{x}} f(\mathbf{x}; \theta)$ with optimizer $\mathbf{x}^*(\theta)$:

$$\frac{dV}{d\theta} = \frac{\partial f}{\partial \theta} \Big|_{\mathbf{x}=\mathbf{x}^*(\theta)}.$$

Proof. By chain rule:

$$\frac{dV}{d\theta} = \underbrace{\nabla_{\mathbf{x}} f \cdot \frac{d\mathbf{x}^*}{d\theta}}_{=0 \text{ by FOC}} + \frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial \theta}.$$

Intuition. First-order gains from adjusting $\mathbf{x}$ vanish at optimum. Only direct effects of θ on f matter.

4.4 Exercise: Profit function Together

Exercise. For $\pi^*(p) = \max_{q \geq 0} pq - C(q)$, find $\frac{d\pi^*}{dp}$.

Solution by envelope.

- $\frac{d\pi^*}{dp} = \frac{\partial(pq - C(q))}{\partial p} \Big|_{q^*} = q^*(p)$
- The supply function!

Verify explicitly.

- FOC: $p = C'(q^*)$ defines $q^*(p)$
- $\pi^*(p) = pq^*(p) - C(q^*(p))$
- $\frac{d\pi^*}{dp} = q^* + p \frac{dq^*}{dp} - C'(q^*) \frac{dq^*}{dp} = q^* + \underbrace{(p - C'(q^*))}_{=0} \frac{dq^*}{dp} = q^*$

Envelope gives this directly without computing $\frac{dq^*}{dp}$.

4.4 Constrained envelope theorem

With constraints and Lagrangian $\mathcal{L} = f + \sum \mu_k h_k$:

$$\frac{dV}{d\theta} = \frac{\partial \mathcal{L}}{\partial \theta} \Big|_{(\mathbf{x}^*, \boldsymbol{\mu}^*)}.$$

Proof. Differentiate $V(\theta) = \mathcal{L}(\mathbf{x}^*(\theta), \boldsymbol{\mu}^*(\theta); \theta)$:

- Terms with $\partial \mathbf{x}^* / \partial \theta$ vanish by stationarity ($\nabla_{\mathbf{x}} \mathcal{L} = 0$)
- Terms with $\partial \mu_k / \partial \theta$ vanish by constraints ($h_k = 0$)

Shadow prices absorb the effect of constraint relaxation.

4.4 Exercise: Indirect utility Together

Exercise. For $V(p, w) = \max u(\mathbf{x})$ s.t. $p \cdot \mathbf{x} \leq w$, find $\partial V / \partial w$.

Solution.

- Lagrangian: $\mathcal{L} = u(\mathbf{x}) + \lambda(w - p \cdot \mathbf{x})$
- By envelope: $\frac{\partial V}{\partial w} = \frac{\partial \mathcal{L}}{\partial w} = \lambda^*$

The Lagrange multiplier on the budget constraint equals the **marginal utility of income**.

This is why λ is often called the shadow price of the budget constraint.

4.4 Exercise: Roy's identity Together

Exercise. Show that $\frac{\partial V / \partial p_i}{\partial V / \partial w} = -x_i^*(p, w)$.

Solution. By envelope:

- $\frac{\partial V}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i} = -\lambda^* x_i^*$
- $\frac{\partial V}{\partial w} = \lambda^*$

Therefore:

$$\frac{\partial V / \partial p_i}{\partial V / \partial w} = \frac{-\lambda^* x_i^*}{\lambda^*} = -x_i^*.$$

This is **Roy's identity**: Marshallian demands can be recovered from indirect utility without solving the consumer problem explicitly.

4.4 Exercise: Cost function envelope Take home

Exercise. For $C(r, \bar{y}) = \min_{\mathbf{z}} r \cdot \mathbf{z}$ s.t. $F(\mathbf{z}) \geq \bar{y}$, find $\partial C / \partial \bar{y}$.

Solution.

- Lagrangian: $\mathcal{L} = r \cdot \mathbf{z} + \lambda(\bar{y} - F(\mathbf{z}))$
- By envelope: $\frac{\partial C}{\partial \bar{y}} = \frac{\partial \mathcal{L}}{\partial \bar{y}} = \lambda^*$

The multiplier equals marginal cost — the cost of producing one more unit of output.

This connects to the firm's supply decision: produce where price equals marginal cost.

4.5 Problems with inequality constraints

Problem. Maximize $f(\mathbf{x})$ subject to $g_j(\mathbf{x}) \leq 0$ for $j = 1, \dots, m$.

Binding vs. slack constraints.

- **Binding (active):** $g_j(\mathbf{x}^*) = 0$ — constraint holds with equality
- **Slack (inactive):** $g_j(\mathbf{x}^*) < 0$ — constraint not tight

Only binding constraints matter for local optimality.

Slack constraints can be ignored locally.

4.5 Feasible directions and the cone

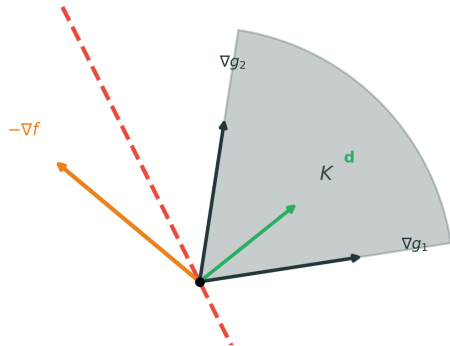
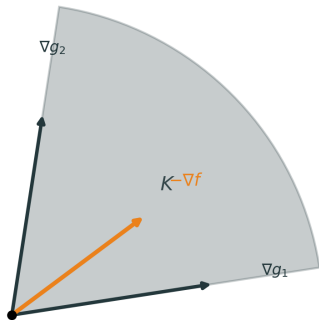
At a boundary point with active constraints, **feasible directions** form a convex cone:

$$D = \{\mathbf{d} : \nabla g_j(\mathbf{x}^*) \cdot \mathbf{d} \leq 0 \text{ for all active } j\}.$$

A direction $\mathbf{d}$ is **improving** if $\nabla f \cdot \mathbf{d} > 0$ (increases objective).

For a local maximum: no feasible direction can be improving.

4.5 Geometric intuition of Farkas lemma



4.5 Statement of Farkas lemma

Farkas Lemma (purely geometric).

Let K be a closed convex cone and $\mathbf{y}$ a vector. Either:

1. $\mathbf{y} \in K$: the vector lies in the cone
2. $\mathbf{y} \notin K$: there exists a separating hyperplane with normal $\mathbf{d}$ such that
 - $\mathbf{k} \cdot \mathbf{d} \leq 0$ for all $\mathbf{k} \in K$ (cone on one side)
 - $\mathbf{y} \cdot \mathbf{d} > 0$ (vector on the other side)

The hyperplane strictly separates the cone from the vector.

4.5 Applying Farkas to optimization

For inequality-constrained optimization:

- Let $K = \text{cone}\{\nabla g_j : g_j \text{ active}\}$ be the cone spanned by active constraint gradients
- Let $\mathbf{y} = -\nabla f$ (negative objective gradient)

Case 1: $-\nabla f \in K$ means $-\nabla f = \sum_j \lambda_j \nabla g_j$ with $\lambda_j \geq 0$.

Case 2: $-\nabla f \notin K$ means there exists feasible improving direction $\mathbf{d}$ where:

- $\nabla g_j \cdot \mathbf{d} \leq 0$ for all j (feasible: doesn't violate constraints)
- $\nabla f \cdot \mathbf{d} > 0$ (improving: increases objective)

4.5 Inequality problem with multipliers

At an optimum, Case 2 is ruled out (no improving feasible direction).

Therefore $-\nabla f \in K$, meaning:

$$\nabla f(\mathbf{x}^*) + \sum_{\text{active}} \lambda_j \nabla g_j(\mathbf{x}^*) = \mathbf{0}, \quad \lambda_j \geq 0.$$

Binding constraints: $\lambda_j > 0$ possible — these are the bottlenecks.

Slack constraints: $\lambda_j = 0$ (they don't contribute).

Geometric interpretation: $-\nabla f$ must point into the feasible cone K spanned by active constraint gradients.

4.5 KKT conditions

Theorem (KKT necessary conditions). At a local maximum (with constraint qualification), there exist multipliers λ_j such that:

1. **Stationarity:** $\nabla f(\mathbf{x}^*) + \sum_j \lambda_j \nabla g_j(\mathbf{x}^*) = \mathbf{0}$
2. **Primal feasibility:** $g_j(\mathbf{x}^*) \leq 0$ for all j
3. **Dual feasibility:** $\lambda_j \geq 0$ (for $g_j \leq 0$, max problem)
4. **Complementary slackness:** $\lambda_j g_j(\mathbf{x}^*) = 0$

CS says: multiplier is zero for slack constraints ($g_j < 0 \Rightarrow \lambda_j = 0$).

Stationarity says: ∇f is a nonnegative combination of active constraint gradients.

4.5 Sign convention summary

With $\mathcal{L} = f + \sum_j \lambda_j g_j$:

Problem	Constraint	λ sign
$\max f$	$g \leq 0$	$\lambda \geq 0$
$\max f$	$g \geq 0$	$\lambda \leq 0$
$\min f$	$g \leq 0$	$\lambda \leq 0$
$\min f$	$g \geq 0$	$\lambda \geq 0$

Memory rule: For max, constraint and multiplier have opposite signs.

$\leq$ constraint $\Rightarrow \lambda \geq 0$.

4.5 Exercise: Budget with nonnegativity Together

Exercise. Maximize $u(x_1, x_2)$ subject to $x_1 + x_2 \leq 1$, $x_1 \geq 0$, $x_2 \geq 0$.

Solution.

- Write: $g_1 = x_1 + x_2 - 1 \leq 0$, $g_2 = -x_1 \leq 0$, $g_3 = -x_2 \leq 0$
- Lagrangian: $\mathcal{L} = u + \lambda_1(1 - x_1 - x_2) + \lambda_2x_1 + \lambda_3x_2$
- All $\lambda_j \geq 0$ for max problem
- Stationarity: $\frac{\partial u}{\partial x_1} - \lambda_1 + \lambda_2 = 0$, $\frac{\partial u}{\partial x_2} - \lambda_1 + \lambda_3 = 0$
- CS: $\lambda_1(1 - x_1 - x_2) = 0$, $\lambda_2x_1 = 0$, $\lambda_3x_2 = 0$

4.5 Exercise: Corner solution analysis Together

Exercise. For $u(x_1, x_2) = x_1 x_2$, determine when the corner $x_1 = 0, x_2 = 1$ is optimal.

Solution.

- At $(0, 1)$: g_1 binds ($\lambda_1 \geq 0$), g_2 binds ($\lambda_2 \geq 0$), g_3 slack ($\lambda_3 = 0$)
- FOC for x_1 : $x_2 - \lambda_1 + \lambda_2 = 0$ gives $1 - \lambda_1 + \lambda_2 = 0$
- FOC for x_2 : $x_1 - \lambda_1 + \lambda_3 = 0$ gives $\lambda_1 = 0$
- But then $\lambda_2 = -1 < 0$, violating dual feasibility

Therefore $(0, 1)$ is **not** optimal. The interior solution is preferred.

Exercise. Minimize $r \cdot z$ subject to $z_1 + z_2 \geq \bar{z}$.

Solution.

- For min with $\geq$ constraint: write $g = \bar{z} - z_1 - z_2 \leq 0$ so $\lambda \geq 0$
- Lagrangian: $\mathcal{L} = r_1 z_1 + r_2 z_2 + \lambda(\bar{z} - z_1 - z_2)$
- FOCs: $r_1 = \lambda$, $r_2 = \lambda$

This implies $r_1 = r_2$.

If input prices differ, only use the cheaper input (corner solution).

Geometric: if $r_1 < r_2$, set $z_2 = 0$, $z_1 = \bar{z}$, and $\lambda = r_1$.

4.6 The key difference: two-sided vs. one-sided

Equality $h(\mathbf{x}) = 0$ allows movement in **both directions** along the constraint surface.

Inequality $g(\mathbf{x}) \leq 0$ restricts to one side: only movement into the feasible region is allowed.

This directional asymmetry explains why:

- Inequality multipliers need sign restrictions ($\lambda \geq 0$)
- Equality multipliers are unrestricted ($\mu \in \mathbb{R}$)

4.6 Formal explanation

Equality: Feasible directions satisfy $\nabla h \cdot \mathbf{d} = 0$.

- Both $\mathbf{d}$ and $-\mathbf{d}$ are feasible
- No sign restriction needed on μ

Inequality: Feasible directions satisfy $\nabla g \cdot \mathbf{d} \leq 0$.

- Only inward directions allowed
- Need $\lambda \geq 0$ to ensure $-\nabla f$ points toward feasible region

Intuition: Equality lets you wiggle both ways; inequality only one way.

4.6 Example: Comparing equality vs. inequality

Example. Consider $\max f(x, y) = xy$ subject to:

Case 1: $x + y = 1$ (equality)

- Can move toward $(0, 1)$ or $(1, 0)$ along the line
- Optimum at $(0.5, 0.5)$ with $\mu = 0.5$ (unrestricted sign)

Case 2: $x + y \leq 1$ (inequality) with $x, y \geq 0$

- Can only move inward (decrease x and/or y) if on boundary
- Same optimum, but now $\lambda \geq 0$ ensures no improvement possible
- Geometric: $-\nabla f$ must point into feasible cone

4.6 The combined Lagrangian

For problems with both equalities and inequalities:

$$\mathcal{L} = f + \underbrace{\sum_k \mu_k h_k}_{\text{equality}} + \underbrace{\sum_j \lambda_j g_j}_{\text{inequality}}.$$

Sign convention:

Multiplier	Sign rule
Equality μ_k	unrestricted (explained in this section)
Inequality λ_j	$\lambda_j \geq 0$ if $g_j \leq 0$; $\lambda_j \leq 0$ if $g_j \geq 0$

Short rule: $\leq$ constraint $\Rightarrow \lambda \geq 0$ (for max); $\geq$ constraint $\Rightarrow \lambda \leq 0$.

4.6 Exercise: Comparing equality vs. inequality Together

Exercise. Maximize $f(x, y) = x^2 - y$ subject to $x^2 + y^2 = 1$ (equality).

Solution.

- Lagrangian: $\mathcal{L} = x^2 - y + \mu(1 - x^2 - y^2)$
- FOCs: $2x - 2\mu x = 0$, $-1 - 2\mu y = 0$
- From first: $x = 0$ or $\mu = 1$
- If $\mu = 1$: $y = -0.5$, $x = \pm\sqrt{0.75}$

Here $\mu = 1 > 0$, but could be negative for different objectives.

The unrestricted μ allows both increasing and decreasing the objective by moving along the circle.

4.6 Exercise: Unrestricted in action Take home

Exercise. Minimize $f(x) = x^2$ subject to $x = 1$ vs. $x \geq 1$. Compare multipliers.

Solution.

Equality $x = 1$:

- $\mathcal{L} = x^2 + \mu(1 - x)$
- FOC: $2x - \mu = 0$, so $\mu = 2$ at $x = 1$

Inequality $x \geq 1$ (i.e., $1 - x \leq 0$):

- For min with $\leq$ constraint: $\lambda \leq 0$
- $\mathcal{L} = x^2 + \lambda(1 - x)$
- FOC: $2x - \lambda = 0$, so $\lambda = 2$, but need $\lambda \leq 0$
- Contradiction! Minimum is at boundary $x = 1$ with $\lambda = 2$ — actually requires re-checking signs.

Appendix — Additional proofs and extensions

This appendix contains:

- A.1 Rigorous proof: concave $\Rightarrow f'' \leq 0$ (Taylor expansion)
- A.2 Log-concavity proofs
- A.3 Multivariate SOC via Hessian definiteness
- A.4 Additional exercises with solutions

Material marked **for interest only**.

A.1 Rigorous proof: concave $\Rightarrow f'' \leq 0$

for interest only

Theorem. If f is concave and $f \in C^2$, then $f''(x) \leq 0$ for all x .

Rigorous proof using Taylor expansion with remainder.

From concavity: $f(x+h) + f(x-h) \leq 2f(x)$ for all $h > 0$.

Taylor expansions around x :

$$\begin{aligned}f(x+h) &= f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \frac{f'''(\xi_1)}{6}h^3 \\f(x-h) &= f(x) - f'(x)h + \frac{1}{2}f''(x)h^2 - \frac{f'''(\xi_2)}{6}h^3\end{aligned}$$

Adding:

$$f(x+h) + f(x-h) = 2f(x) + f''(x)h^2 + \frac{f'''(\xi_1) - f'''(\xi_2)}{6}h^3.$$

By concavity: $f''(x)h^2 + \frac{f'''(\xi_1) - f'''(\xi_2)}{6}h^3 \leq 0$.

Divide by h^2 and let $h \rightarrow 0$: third order terms vanish, leaving $f''(x) \leq 0$.

A.2 Concave positive implies log-concave

for interest only

Theorem 1. Concave and positive $\Rightarrow$ log-concave.

Proof. If $f > 0$ and f concave, then $\ln f$ is concave.

For $\lambda \in [0, 1]$:

$$\ln f(\lambda x + (1 - \lambda)y) \geq \ln [\lambda f(x) + (1 - \lambda)f(y)]$$

by concavity of f and monotonicity of $\ln$. Since $\ln$ is concave:

$$\ln [\lambda f(x) + (1 - \lambda)f(y)] \geq \lambda \ln f(x) + (1 - \lambda) \ln f(y).$$

Thus $\ln f(\lambda x + (1 - \lambda)y) \geq \lambda \ln f(x) + (1 - \lambda) \ln f(y)$.

A.2 Log-concave implies quasiconcave

for interest only

Theorem 2. Log-concave $\Rightarrow$ quasiconcave.

Proof. If $\ln f$ concave, then $f = e^{\ln f}$ is quasiconcave (composition of increasing concave with concave).

Specifically: if $\ln f(\lambda x + (1 - \lambda)y) \geq \lambda \ln f(x) + (1 - \lambda) \ln f(y)$, then

$$f(\lambda x + (1 - \lambda)y) \geq f(x)^\lambda f(y)^{1-\lambda} \geq \min\{f(x), f(y)\}.$$

A.3 Multivariate SOC: Hessian definiteness

for interest only

For $f \in C^2$: Taylor at critical point $\mathbf{x}^*$:

$$f(\mathbf{x}^* + \mathbf{h}) = f(\mathbf{x}^*) + \frac{1}{2} \mathbf{h}^\top H_f(\mathbf{x}^*) \mathbf{h} + o(\|\mathbf{h}\|^2).$$

Theorem. If $H_f(\mathbf{x}^*)$ is negative definite (all eigenvalues < 0), then $\mathbf{x}^*$ is a strict local maximum.

Proof. For small $\mathbf{h} \neq 0$, the quadratic form dominates the remainder.

Negative definiteness ensures $f(\mathbf{x}^* + \mathbf{h}) < f(\mathbf{x}^*)$.

Connection to concavity: f concave $\Leftrightarrow H_f(\mathbf{x})$ negative semidefinite for all $\mathbf{x}$.

A.4 Additional exercise: Multivariate critical points

for interest only

Exercise. Find critical points of $f(x, y) = x^3 + y^3 - 3xy$.

Solution.

- $\nabla f = (3x^2 - 3y, 3y^2 - 3x) = (0, 0)$
- From first: $y = x^2$
- Substitute: $x^4 - x = 0$, so $x(x^3 - 1) = 0$
- Solutions: $(0, 0)$ and $(1, 1)$

Hessian at $(1, 1)$: $H = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$, eigenvalues $3, 9 > 0$ (local min).

At $(0, 0)$: $H = \begin{bmatrix} 0 & -3 \\ -3 & 0 \end{bmatrix}$, determinant $-9 < 0$ (saddle).

A.4 Additional exercise: Borderline SOC

for interest only

Exercise. Analyze $f(x) = -x^4$ at $x = 0$ where SOC with $f'' < 0$ fails.

Solution.

- $f'(0) = 0$, $f''(0) = 0$, $f'''(0) = 0$, $f^{(4)}(0) = -24 < 0$

Fourth-order Taylor: $f(h) = -h^4 < 0 = f(0)$ for $h \neq 0$.

Strict local maximum.

This shows higher-order terms can determine optimality when SOC is inconclusive.